User Inputs-The Prompt() Function

Write javascript programs to solve all of the following problems. Some of them will be partially solved. Fill in the blanks to make it work.

Exercise : Input Hours to Seconds

Write a program that takes in a number that represents hours and converts it to seconds. The program should return the number of seconds.

```
let hours = +prompt("Enter the number of hours:");

let seconds = hours * 3600;

console.log("That is " + seconds + " seconds.");
```

Exercise : Input Area of a Triangle

Write a program that takes two numbers that represent the base and the height of a triangle and returns the area of the triangle.

```
let base = +prompt("Enter the base of the triangle:");

let height = +prompt("Enter the height of the triangle:");

let area = (base * height) / 2;

console.log("The area of the triangle is " + area);
```

Input Exercise : ERA

A pitcher's ERA (earned runs average) is calculated by multiplying the earned runs times total innings in the game (usually 9) and then dividing by the total innings that the pitcher pitched.

Write a program that assumes that there were 9 innings in the game and takes in two numbers that represent the earned runs and the innings pitched. The function should return the ERA.

```
let earnedRuns = +prompt("Enter the number of earned runs:");

let inningsPitched = +prompt("Enter the number of innings pitched:");
```

```
let ERA = (earnedRuns * 9) / inningsPitched;

console.log("The ERA is " + ERA);
```

Input Exercise : Area of a Circle w/ Radius

Write a program that displays the area of a circle given the radius.

Use the constant `Math.PI` from the `Math` class.

```
let radius=+prompt("What is the radius of a circle whose area you want me to calculate?");

let area=__Math.PI*radius_______*radius**2;

console.log("The area is " + area + " cm squared.");
```

Input Exercise : Square a Double Type

Write a program that computes the square of the number entered by a user.

```
let number=_+__prompt("Enter a number. I will give you the square of it.");

let square=number**2;

console.log("The square of the number is " + ____square_____);
```

Input Exercise : Cube of a Number

Write a program that computes the cube of the number provided by the user.

```
let num = +prompt("Enter a number. I will give you the cube of it.");

let cube = num ** 3;

console.log("The cube of the number is " + cube);
```

Input Exercise : Weight of a Car Carrier Trailer

Write a program that returns how much weight is on a car carrier trailer given the average weight of a car and number of cars on the trailer.

```
let carWeight=+prompt("Enter the average weight of a car in lbs.")

let numberOfCars=+prompt("How many of these cars will be placed on the trailer.")

let totalWeight=carWeight* numberOfCars;

console.log("The total weight of all cars on the trailer is "+totalWeight);
```

Input Exercise : Purchase Calculator

Create a simple calculator that combines four numbers to determine the pre-tax amount of 4 purchased items at a store.

```
let item1 = +prompt("Enter price of item 1:");

let item2 = +prompt("Enter price of item 2:");

let item3 = +prompt("Enter price of item 3:");

let item4 = +prompt("Enter price of item 4:");

let total = item1 + item2 + item3 + item4;

console.log("The total cost before tax is $" + total);
```

Input Exercise : Product

Write a program that displays the product of two numbers entered by a user.

```
let num1 = +prompt("Enter the first number:");

let num2 = +prompt("Enter the second number:");

let product = num1 * num2;

console.log("The product of the two numbers is " + product);
```

Input Exercise : Average

Write a program that displays the average of 4 given numbers!

```
let number1=+prompt("Enter the first number:");
let number2=+prompt("Enter the second number:");
let number3=+prompt("Enter the _third_____ number:");
let _number4_____=+prompt("Enter the fourth number:");
let average=(number1+number2+number3+number4)/_____4____;
console.log("The average of these numbers i s "+average);
```

Input Exercise : Circumference of Circle

Write a program that returns the circumference of a circle given the radius! Assume pi is 3.14.

Note: the equation for circumference is `2pi*radius`.

```
let radiusC = +prompt("Enter the radius of the circle:");
let circumference = 2 * 3.14 * radiusC;
console.log("The circumference of the circle is " + circumference);
```

Input Exercise : Volume of Sphere

Write a program that returns the volume of a sphere given the radius. Assume pi is 3.14.

Note: The equation for a sphere's volume is (4/3)pi * $radius^3$

```
let radiusS = +prompt("Enter the radius of the sphere:");
let volume = (4 / 3) * 3.14 * radiusS ** 3;
console.log("The volume of the sphere is " + volume);
```

Input Exercise : Pythagorean

Write a program that takes in two numbers that represent the legs of a triangle and returns the length of the hypotenuse by using the Pythagorean theorem.

The Pythagorean Theorem is a2 + b2 = c2.

```
let a = +prompt("Enter the length of side a:");
let b = +prompt("Enter the length of side b:");
let c = Math.sqrt(a ** 2 + b ** 2);
console.log("The length of the hypotenuse is " + c);
```